Creating dashboards

For manual creation, click on *Create (plus icon)* > *Dashboard* > *Add new panel*. Write queries (e.g., PromQL for Prometheus) and visualise data (graphs, gauges, tables). Save the dashboard.

You can also import pre-built official/community dashboards from Grafana Dashboards. Click on *Import* and paste the dashboard ID (e.g., 1860 for Node Exporter).

Setting up alerts

To create alert rules, edit a panel and go to *Alert* > *Create Alert*. Define conditions (e.g., CPU usage > 90%). Configure notification channels by going to email/Slack/PagerDuty and clicking on *Alerting* > *Notification* channels. Add contact points (e.g., Slack webhook).

Securing Grafana

Change the default password by clicking on *Admin* > *Profile* > *Change Password.*

Enable HTTPS. Edit Grafana *config* file (*/etc/grafana/grafana.ini*) as follows:

```
ini
[server]
protocol = https
cert_file = /path/to/cert.pem
cert_key = /path/to/key.pem
Restart Grafana:
sudo systemctl restart grafana-server
```

Grafana enables comprehensive visualisation and monitoring of data in real-time. Users can set up data sources, design dashboards, and configure alerts through the web interface. Moreover, the guided walkthrough from installation to integration and dashboard setup facilitates the monitoring of system metrics, business KPIs, and other time-series data.

Unified observability across complex IT infrastructures is made possible with Grafana due to its broad support of data sources, which include Prometheus, InfluxDB, MySQL, and Elasticsearch. Furthermore, its agile and strong plugin ecosystem enhances the capability to be tailored to an array of use cases in DevOps, IoT, analytics, and business intelligence.

Properly configuring a Grafana environment greatly enhances operational and system insights, facilitating better decision-making while boosting the reliability of the system. This underlines its critical importance in contemporary monitoring and observability frameworks. END

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